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DENSYS 2.0

ERASMUS MUNDUS DECENTRALISED SMART ENERGY SYSTEMS



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ENERGY ECONOMICS ELECTIVE

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ENERGY ECONOMICS ASSIGNMENT

Class Exercise 2:

QUESTION: A new small cogeneration plant installation is expected to reduce a company's annual energy bill by 486,000 €. If the capital cost of the new boiler installation is 2,220,000 € and the annual maintenance and operating costs are 42,000 €, determine the expected payback period for the project.

ANSWER

Net Annual Savings = Annual energy savings – Annual O&M cost

Net Annual Savings = 486,000 € - 42,000 €

Net Annual Savings = 444,000 €

The Payback period is the time required for the cumulative net savings to equal the initial capital investment.

$$\begin{aligned} \text{Payback Period} &= \frac{\text{Initial Capital Cost}}{\text{Net Annual Savings}} \\ &= \frac{2,200,000\text{€}}{444,000\text{€}} \end{aligned}$$

Payback period= 5 years

The expected payback period for the cogeneration plant project is 5 years.

Class Exercise 3:

Suppose that a power plant costs 10 billion € to build and has an expected life of 30 years. The variable cost of producing one MWh of electricity is 20 €. It will operate 24 hours a day, 360 days a year at a capacity of 1000 MW. What is the levelized cost of energy if the interest rate is 5%?

ANSWER

Annual operating hours= 24h/day x 360days/year= 8640h/year

Annual energy production= 1000MW x 8640h = 8,640,000MW

We then find the annualized capital cost using the Capital Recovery Factor (CRF) formula.

$$CRF = \frac{r(1+r)^n}{(1+r)^n - 1}$$

Where-

r = interest rate = 5% = 0.05

n= number of time period= 30 years

$$CRF = \frac{0.05(1.05)^{30}}{(1.05)^{30} - 1}$$

$$CRF = 0.065$$

Annualized capital cost = Capital Cost x CRF

$$= 10,000,000,000 \text{ €} \times 0.065$$

Annualized capital cost = 650,000,000/year

We then proceed to find the annual variable cost

The Annual variable cost = Annual energy production x Variable cost per MWh

$$= 8,640,000 \text{ MWh} \times 20 \text{ €/MWh}$$

Annual variable cost = 172,800,000€/year

We then calculate the annual cost

Total annual cost = Annualized capital cost + Annual variable cost

$$= 650,000,000 + 172,800,000 \text{ €}$$

Total annual cost = 822,800,000/year

We then calculate the Levelized Cost of Energy (LCOE)

$$LCOE = \frac{\text{Total Annual Cost}}{\text{Annual Energy Production}}$$

$$LCOE = \frac{822,800,000}{8,640,000}$$

$$LCOE = 95.23 \text{ €/MWh}$$

Therefore, the Levelized cost of energy for the power plant is 95.23€/MWh

ENERGY MARKETS TD1

Exercise III: Spot Market: supply and demand curves

B. Consider a market that has 5 generators. For a given time period (3pm and 4pm the following day), the market operator has received a set of single-hourly supply offers from these 5 participants. These offers are defined in tables 1 and 2.

Table 1. Available Energy and Prices for different technologies

Technology supplied	Short name	Quantity (MWh)	Price (€)
Gas	S1	15	75
Nuclear	S2	100	15
PV	S3	32	0
CHP	S4	25	42
Hydro	S5	70	10

Table 2. Energy demand and associated prices

Demand	Quantity (MWh)	Price (€)
D1	35	65
D2	23	78
D3	12	10
D4	38	46
D5	43	63
D6	16	32
D7	57	50

QUESTION 1

Give the definition of the supply/demand curve in the day-ahead market

Answer

Supply curve (day-ahead market): The supply curve represents the cumulative quantity of electricity that generators are willing to supply at different price levels, arranged in ascending order of price (merit order). It's a step function where each step represents a generator's offer. Each generator offers a quantity of energy (MWh) at a specific price (€/MWh). When stacked cumulatively from the cheapest to the most expensive unit, this forms a non-decreasing stepwise curve.

Demand curve (day-ahead market): The demand curve represents the cumulative quantity of electricity that consumers are willing to purchase at different price levels, arranged in descending order of price (willingness to pay). It's also a step function where each step represents a demand bid. Each consumer bids a quantity at a maximum price they are willing to pay. When stacked cumulatively from the highest to the lowest price, this forms a non-increasing stepwise curve.

Market-clearing point is at the intersection of these two curves: all supply offers with bid price at or below this price, and all demand bids with price at or above it, are accepted. Simply put, it is where cumulative supply equals cumulative demand.

QUESTION 2: Make an Excel plot of the supply/demand curve for this market time unit.

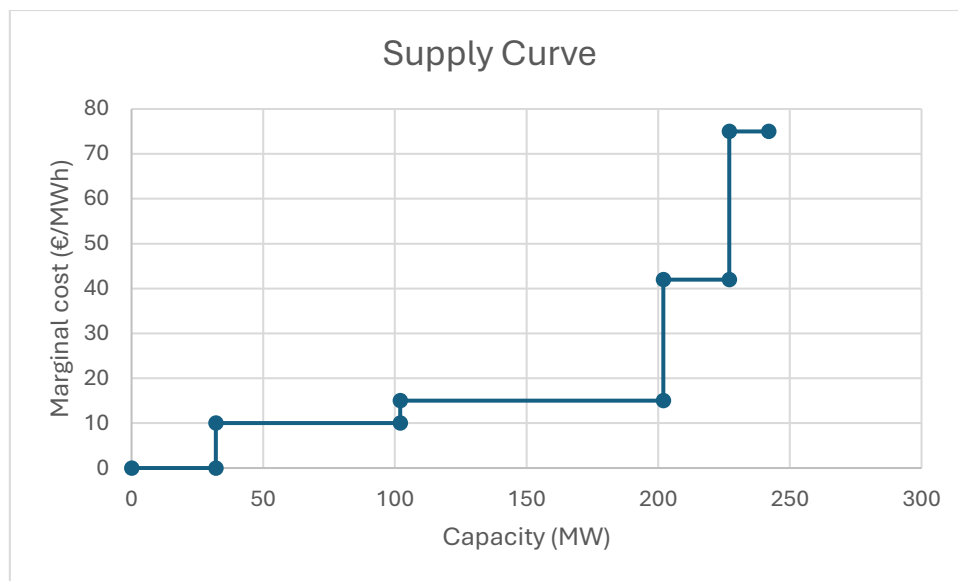


Figure 1: Supply Curve

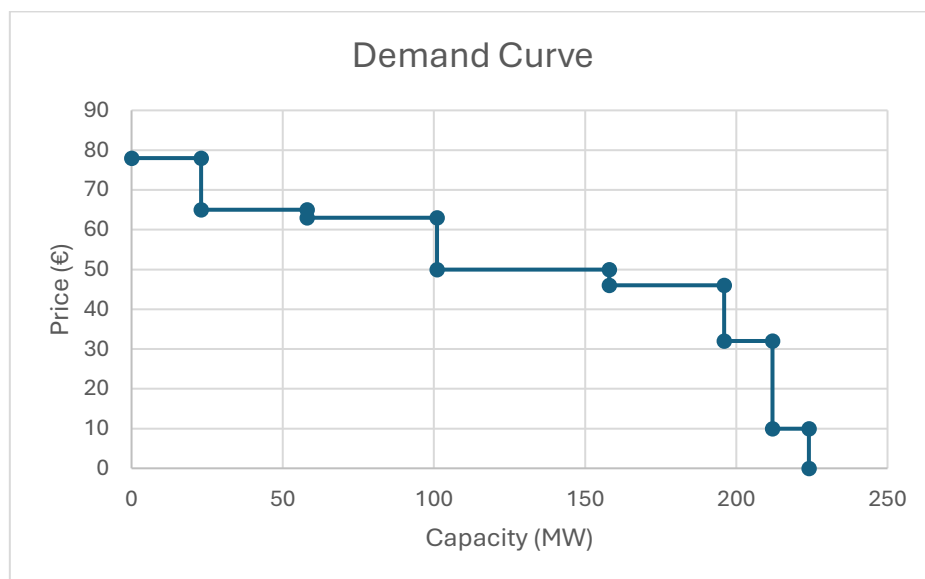


Figure 2: Demand Curve

QUESTION 3: Compute the total amount of energy supplied/asked for through the market? Compare them. Is there a problem?

Total supply (from our supply table in Excel)

$$= 32\text{MWh} + 70\text{MWh} + 100\text{MWh} + 25\text{MWh} + 15\text{MWh}$$

$$= 242 \text{ MWh}$$

Total demand (from our demand table in Excel)

$$= 35\text{MWh} + 23\text{MWh} + 12\text{MWh} + 38\text{MWh} + 43\text{MWh} + 16\text{MWh} + 57\text{MWh}$$

$$= 224 \text{ MWh}$$

Comparison: Supply (242 MWh) > Demand (224 MWh)

Is there a problem?

No, there is no problem. The total supply available (242 MWh) exceeds the total demand requested (224 MWh), which is normal and expected in electricity markets.

The market operates on a merit order principle where suppliers offer their capacity at different prices, but the market only dispatches the cheapest generators needed to meet demand at the equilibrium price.

The surplus of supply over demand is beneficial for several reasons. First, it ensures competition among suppliers, which keeps prices competitive. Second, it provides reserve capacity that serves as backup power available if needed, such as in balancing markets when unexpected shortages occur. Third, it promotes market efficiency by ensuring only cost-effective generators are dispatched. Finally, it enhances system reliability by providing the ability to handle unexpected demand increases or supply issues. The more expensive generators (CHP at €42/MWh and Gas at €75/MWh) remain available but are not dispatched in the day-ahead market, demonstrating that the market is functioning properly by selecting the most economical generation mix to meet demand.

QUESTION 4: Cite other types of offers than single-hourly offers

- **Block bids:** offers for fixed quantities over several (or multiple) consecutive hours that must be accepted or rejected as a whole (e.g. 50 MWh in each of hours 8–12).
- **Complex bids:** offers that include technical constraints such as start-up cost, minimum run time or ramping limits, which require unit-commitment style optimization rather than simple merit-order sorting
- **Linked or conditional bids:** here, acceptance of one bid may depend on acceptance or rejection of another (e.g. only run plant B if plant A is also accepted).
- **Flexible offers-** Quantity can be produced in any hour within a time window. Can vary price/quantity by hour
- **Balancing reserve offers-** Used in intraday and balancing markets (upward/downward regulation)

QUESTION 5: Plot the supply and demand curves on the same plot.

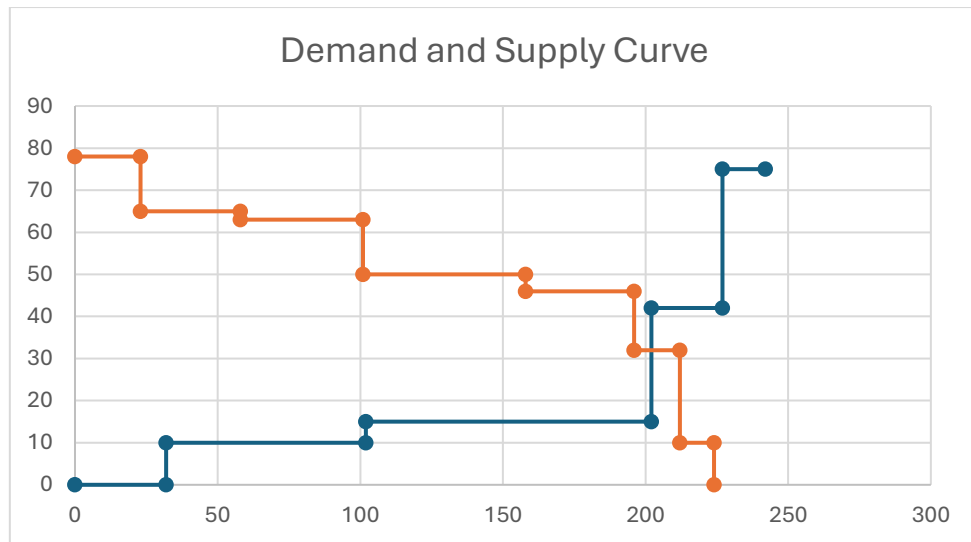


Figure 3: Supply and Demand Curve

QUESTION 6: What is the market clearing price and quantity? Who is effectively supplying power and how much? Who is effectively consuming and how much?

From the combined supply and demand curves, the intersection occurs at a market clearing price of **32 €/MWh**

Market clearing:

- Clearing price: 32 €/MWh
- Clearing quantity: 202 MWh

Suppliers: Looking at the ordered supply offers

- S3 (PV): offers 32 MWh at 0 €/MWh → Fully accepted, Dispatched: 32 MWh
- S5 (Hydro): offers 70 MWh at 10 €/MWh → Fully accepted, Dispatched: 70 MWh
- S2 (Nuclear): offers 100 MWh at 15 €/MWh → Dispatched: 100 MWh

Total supplied: 202 MWh

S4 (CHP) bids 42 €/MWh and S1 (Gas) bids 75 €/MWh, both above 32 €/MWh, so they are not dispatched in the day-ahead market.

All accepted generators are paid the **uniform market price of 32 €/MWh** for each MWh they produce.

Consumers (willing to pay 32 €/MWh or above): on the demand side,

- D2: 23 MWh at 78 €/MWh → Served: 23 MWh
- D1: 35 MWh at 65 €/MWh → Served: 35 MWh
- D5: 43 MWh at 63 €/MWh → Served: 43 MWh

- D7: 57 MWh at 50 €/MWh → Served: 57 MWh
- D4: 38 MWh at 46 €/MWh → Served: 38 MWh
- D6: 16 MWh at 32 €/MWh → Served: 6 MWh

Total consumed: 202 MWh

D3 bids only 10 €/MWh, below the clearing price, so it is not served in the day-ahead market.

D2, D1, D5, D7, D4 and D6 have bid prices at or above 32 €/MWh and together demand 202 MWh, so all are supplied at 32 €/MWh.

Exercise IV: Intraday and balancing market

Some markets players can provide balancing services for that time unit. These players already participated in the day-ahead electricity market:

QUESTION 7: Draw the supply curve corresponding to these balancing offers.

To construct the supply curve for balancing offers, we organize the available balancing capacity presented in table 3 in merit order (from lowest to highest marginal cost) as shown in table 4.

Table 3. Quantified energy properties for different technologies

Technology supplied	Short name	Upward Quantity (MWh)	Price up (€)	Downward Quantity (MWh)	Price Downward (€)	Cumulative production
Hydro	S5	20	35	20	5	20
CHP	S4	5	50	15	15	25
Gas	S1	15	80	-	-	40

Table 4. Merit Order of Supply for different technologies

Capacity (MW)	Marginal cost (€/MWh)
0	35
20	35
20	50
25	50
25	80
40	80

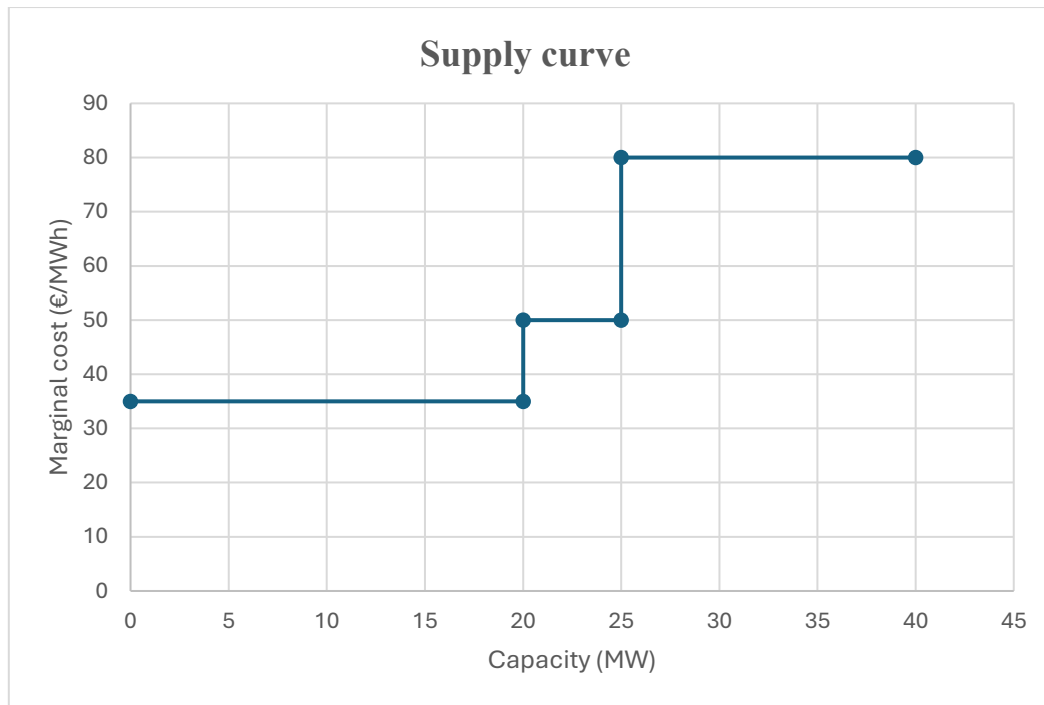


Figure 4: Supply Curve for the Balancing offers

QUESTION 8: The nuclear power plant from the previous exercise has a problem with its cooling system, yielding a shortage of 22MWh in comparison with the amount contracted through the day-ahead market. What is the resulting demand for our balancing market? Add this demand curve to your previous plot.

Answer

It is stated that the nuclear power plant produces 22 MWh less than the amount it committed to deliver in the day-ahead market. This creates a shortage, meaning that total supply is lower than total demand. To restore system balance, the TSO must procure additional energy through the balancing market.

Resulting balancing demand

Type of imbalance: Shortage

Amount of balancing energy required: 22MWh of upward balancing energy (we need other generators to increase their production to fulfil the missing nuclear generation and maintain supply-demand balance)

In the balancing framework, this can be represented as a “demand” for upward regulation equal to 22 MWh: graphically, we represent this with a vertical line at 22 MWh that must intersect the upward balancing supply curve.

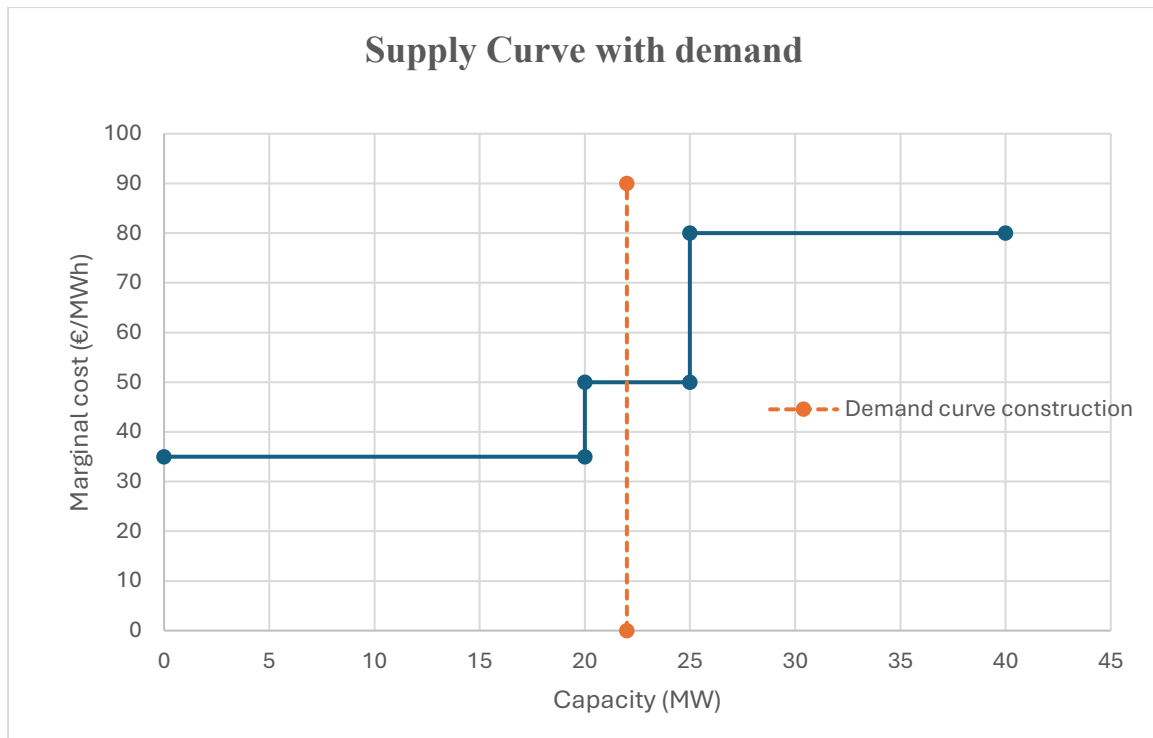


Figure 5: Supply Curve with demand

QUESTION 9: What is the resulting balancing price, and who will provide the balancing service? How many MWh and in which direction?

Answer

The TSO covers the 22 MWh shortage by activating upward balancing offers starting from the lowest price, following the merit-order principle.

Table 5: Balancing market

Generator	Available upward energy	Activated energy	Price
Hydro	20	20	35
CHP	5	2	50
Gas	15	0	80

The cheapest balancing energy is bought first. For the shortage of 22 MWh, from Table1 above, we can see that;

Hydro has 20 MWh available, but we need 22 MWh. So, we use all 20 MWh from Hydro. But we have a remaining shortage of 2MWh

CHP has 5 MWh available, but we need 2 MWh. So, we use only 2 MWh from CHP. Now the shortage is fully covered.

Gas is not needed at all, so it provides 0 MWh

What is the resulting balancing price?

Balancing markets use uniform pricing, meaning that all activated balancing energy is paid at the price of the marginal unit, i.e. the last unit required to cover the imbalance. Everyone is paid the price of the last (most expensive) activated unit and the last unit (marginal unit) used was **CHP at the price of 50 €/MWh**

So: **Balancing price = 50€/MWh**

Who will provide the balancing service? How many MWh and in which direction?

The first 20 MWh are supplied upward by Hydro

The remaining 2 MWh are supplied upward by CHP

Supplied in an Upward direction.

QUESTION 10: Compute the revenues and payments for all participants in this balancing market. Deduce the combined revenues, considering both day-ahead and balancing markets.

Answer

The balancing market is settled at a price of 50 €/MWh as discussed above.

For Hydro: 20MWh x 50 €/MWh = €1,000

For CHP: 2MWh x 50 €/MWh = €100

Total balancing payment: €1,100

Combined revenue:

Day-Ahead Market Revenues and Balancing Market Adjustment:

From the day-ahead market clearing at **€32/MWh** with **202 MWh** total:

Table 6 Combined revenue of the participants of the day-ahead and balancing market

Supplier	Day Ahead			Balancing market adjustment		Total Revenue
	Day Ahead Quantity	Market Price (€32/MWh)	Revenue (€)	Balancing adjustment	Balancing settlement	
PV (S3)	32	32	1024	0	0	1024

Hydro (S5)	70	32	2240	+20MWh (settled at balancing price of €32/MWh)	1000	3240
Nuclear (S2)	100	32	3200	minus balancing charge of the imbalance caused	-1100	2100
CHP (S4)	0	32	0	+ 2MWh (settled at balancing price of €32/MWh)	100	100

Key Observations:

From the combined revenue outcome from the day-ahead and balancing market, we observe a distribution of capacity and therefore revenues from the market participants. This results in ‘winners’ and ‘losers’ in this market.

Winners from balancing market:

- **Hydro (S5)** gains the most. From the day-ahead market, it was dispatched but it is still able to provide an upward balance to cater for the imbalance caused by nuclear in the balancing market. We see that it bids €35/MWh for upward balancing and receives €50/MWh (balancing market clearing price) which is above both its day-ahead and balancing offer price. Therefore, from the table, we observe that it earned an additional €1000 in balancing revenue. Its total revenue resulting to €3240, which is the highest in the market.
- **CHP (S4)** was not dispatched in the day-ahead market, but it was able to provide an upward balance in the balancing market and therefore earns €100 (as illustrated earlier). We observe that the balancing market not only providing balancing services but also incentivizes reliable performance. And in CHP case, we can see that the balancing market creates revenue opportunities for high-cost generators that are too expensive to be dispatched in the day-ahead market.

Losers from balancing market:

- **Nuclear (S2)** loses in this market. Though it was fully dispatched in the day-ahead market, it experiences a shortage of 22MWh. As a result, it pays penalty of €1100 (as calculated above) to compensate for the imbalance caused. We observe that the balancing market provides financial consequences for unreliable generators, thus, creating incentives for better maintenance and forecasting.

INVESTMENT DECISION AND MANAGERIAL FLEXIBILITY IN THE ENERGY SECTOR

Part A

The calculated values for the NPV for plan A and plan B are obtained as 162.15M € and 141.52M € respectively as shown in table 1. Plan A with a higher NPV presents an economic interest in a deterministic (fully known) future.

Table 1. Calculated NPV Values for plan A and plan B

Year	0	1	2	3
Demand forecast (M)		0.3	0.6	0.9
Discount rate (%)		9%		

Plan A				
Number of Production line		3	3	3
Production Capacity (M)		0.9	0.9	0.9
Capex total (M\$)	900	0	0	0
Opex per unit (\$)		1280	1280	1280
Salvage (M\$)				0
Unit selling price (\$)		2000	2000	2000
PV (M)		198.17 €	363.61 €	500.37 €
NPV		162.15 €		

Plan B				
Year	0	1	2	3
Number of Production line		1	2	3
Production Capacity (M)		0.3	0.6	0.9
Capex total (M\$)	300	300	300	0
Opex per unit (\$)		1500	1500	1500
Salvage (M\$)				300
Unit selling price (\$)		2000	2000	2000
PV (M)		-137.61 €	0.00 €	579.14 €
NPV		141.52 €		

Figure 1 shows the sensitivity analysis for the two different projects. At a 9% discount rate, Plan A is the preferred choice due to the higher NPV. The sensitivity analysis examines the effect of the discount rate on the feasibility of Plan B as compared to plan A. At lower discount rates, plan A with a higher upfront investment but a larger long term cash flow achieves higher NPVs. As the discount rate increases, the NPVs of plan A decreases faster than that of plan B with incremental investments. Beyond a particular threshold at a discount rate of approximately 12%, switching occurs and plan B has higher values of NPV meaning the plan/project with the lower CAPEX becomes more favorable.

Plan A represents a large-scale, capital-intensive project e.g. nuclear plant or renewable energy plant with long lifetimes like hydropower having lower operating costs. Plan B represents a modular approach in terms of investment in a plant e.g. a fossil fuel-based plant. A low discount rate corresponds to stable frameworks while a high discount rate corresponds to regulatory uncertainty and capital-intensive projects thrive in this environment. The success of plan A depends on low interests at the start of the project. Plan B relies on forecasted market demand to build capacity. A high discount rate reflects market risks and applies a huge penalty on projects with distant revenues like plan A. This then explains why fossils look more attractive in high-risk situations.

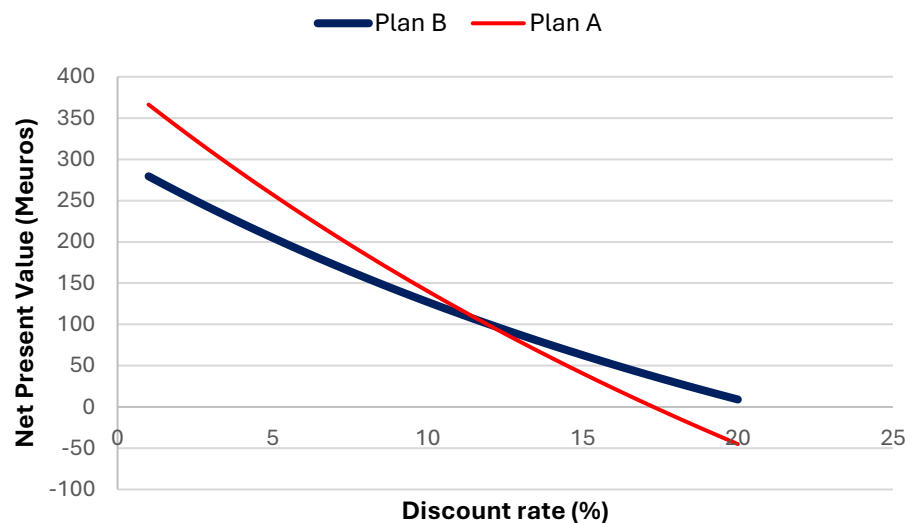


Figure 1. Evolution of NPV with different discount rates

Part B

In Part B, we move from a deterministic scenario to an uncertain energy future. A thousand scenarios are simulated to analyze how volatility in demand impacts the profitability and risk of both investment strategies. Plan A returns significantly higher average values for NPV. This is due to a lower OPEX during scaling and unmatched profits when demands are high. However, this poses a high risk since when the demand is low, one is stuck with a high-capacity plant that isn't generating revenue. For plan B, the NPV is much lower for most instances. This is because it is hindered by its higher OPEX and by the fact that it reaches 0.9M capacity only in its final year. However, the risks involved with plan B are much lower because the investments are spread out.

Plan A, with its high fixed costs and lower variable costs, is like a renewable plant (high CAPEX, low OPEX). Plan B, with higher OPEX, is more like a gas-turbine plant that is cheaper to build but more expensive to run. Part B shows that without flexibility (the ability not to invest), Plan B looks quite weak because it has the worst of both worlds: high variable costs without the full benefit of scale.

Figure 2 represents each plan's NPV distribution. Plan A has a wider, flatter distribution with a broader range of NPVs as observed in table 2. It is highly sensitive to demand. If the demand

is high (right side of the graph), the low OPEX creates massive profit. If demand is low (left side), the high CAPEX creates a financial disaster. Plan B has a peak distribution with averagely lower values of the NPV. Since the investment is spread out, the project is safe even though the return is not as big. Without flexibility, Plan A is the more rational choice for an investor because of the higher mean. However, for a company trying to avoid the risk associated with going all in on initial investment and protecting itself from a dip in demand Plan B might seem more suitable.

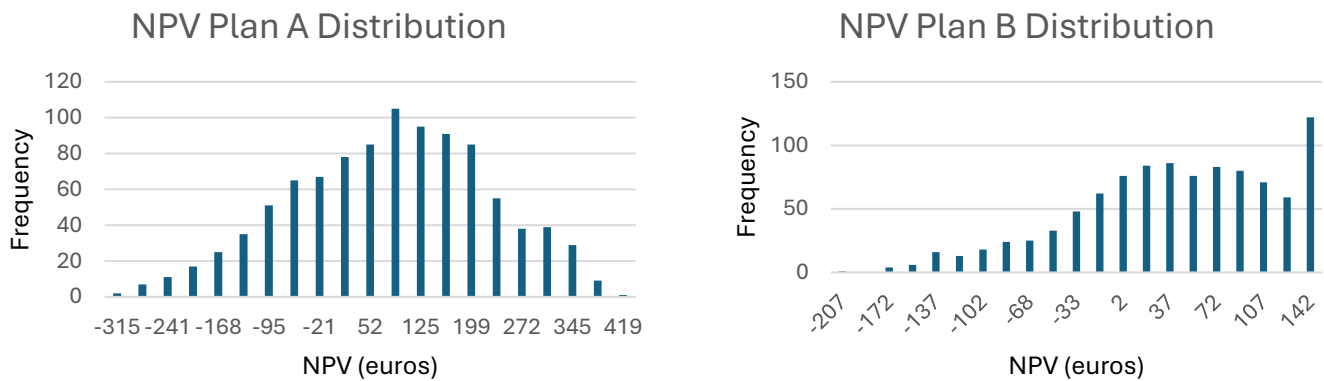


Figure 2. NPV Distribution for Plans A and B for Part B

Table 2. Comparison of statistical data between Plans A and B for Part B at one instance

	Plan A	Plan B
Mean	102.34	49.51
Standard Deviation	144.86	70.68
NPV	301.14	138.83

Part C

In part C, the most critical concept of modern energy concepts is introduced: Managerial Flexibility. While Part B involves investing in a modular way regardless of the demand, a company has the option to wait for a forecast in part C and act accordingly. This shift from a fixed plan to a flexible plan between Part B and C results in financial improvement as shown in tables 3 and 4. The mean values and NPVs for Plan B in Part C are higher than those of plan A and hence the decision is flipped with reference to part C. By choosing a modular approach with the option to expand, the firm achieves a higher expected return than the large-scale plant (Plan A) while maintaining much lower risk.

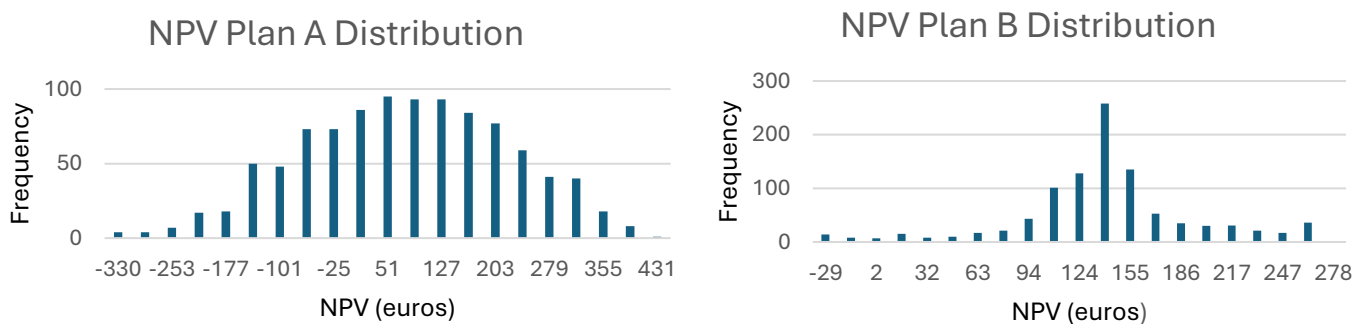
Table 3. Comparison of statistical data between Plans A and B for Part C at one instance

	Plan A	Plan B
Mean	100.7	142.27
Standard Deviation	145.6	55.66
NPV	73.27	263.3

Table 4. Comparison of statistical data between fixed investment (Part B) and flexible investment (Part C) at one instance

	Plan B	
	Part B	Part C
Mean	49.51	142.27
Standard Deviation	70.68	55.66
NPV	138.83	263.3

Figure 3 shows the NPV distributions for plans A and B. An observable difference in the NPV Plan Distribution for Plan B for parts B and C is the truncation of negative values in part C initially evident in part B. This is majorly contributed by the fact that the company decides only to invest when the demand is sufficiently high in part C, hence **capping losses**.

**Figure 3. NPV Distribution for Plans A and B for Part C**

For a green hydrogen production company, one might start with a small electrolyzer and if demand for hydrogen increases or the wind farm producing energy for electrolysis produces more power, then the company can invest in a new electrolyzer otherwise continue producing with the initial electrolyzer. Flexibility in part C also shows avoiding unnecessary investments in a changing economy where new technologies emerge in the energy sector.

Uncertainty analysis

The uncertainty analysis involves maximizing the positive side of the NPVs/ increasing the concentration on the positive side. This can be done through taking the following actions:

1. Reducing the discount.

Reducing the discount makes high demand years e.g. year 3 in this case much valuable hence shifting the NPVs to the positive side.

2. Allowing flexibility

Flexibility modifies how uncertainty affects NPV rather than eliminating uncertainty itself. The option to delay or expand reduces the risks involved with fixed modular expansion and heavy initial investment. It also truncates the negative values as previously observed in the analysis of histograms for Part B and Part C.

3. Reducing Variable Cost (OPEX)

Reducing in OPEX can be achieved by learning during running of a particular plant. The operational costs would then reduce with time and attract larger cash flows in later years.

4. Power Purchase Agreements (Selling Price)

Since the market price cannot be controlled by the company, the company can have signed contractual agreements with the buyer to fix a selling price hence eliminating the uncertainty of market prices. This truncates the negative end of values. However, when prices spike, this mechanism does not benefit the company and benefits are limited.